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Ethanol condensation at elevated pressure over copper on AlMgO and AlCaO porous mixed-oxide supports†

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The coupling of ethanol to produce C₄-compounds by Guerbet condensation at 325 °C was studied using porous mixed oxide catalysts of AlMgO and AlCaO prepared by the emulsion mediated method with and without the addition of Cu. The synthesis procedure was effective in preparing materials with higher porosity and BET area. Cu catalysts show improved yields and selectivities at elevated H₂ pressures, which is related to ethanol dehydrogenation equilibrium shifts. Without Cu, AlMgO catalysts were more selective to diethyl ether (45–55 C%) than higher alcohols and aldehydes (30–40 C%), while AlCaO catalysts had no activity for ethanol conversion. The addition of Cu inhibited production of diethyl ether while promoting dehydrogenation to acetaldehyde, the initial step in Guerbet coupling of ethanol. Copper addition increased the activity and the selectivity to higher alcohols and aldehydes (up to 55 C% selectivity) for all catalysts. Olefins and paraffins comprised the majority of the side products formed with Cu-doped catalysts. The ethanol conversion levels off near 30% with decreasing space velocity due to inhibition by water and other products.

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1. Introduction

Ethanol is currently the world's most produced liquid fuel derived from biomass sources, particularly in the Americas, where the United States consumed over four times more ethanol fuel than biodiesel in 2017.^{1,2} Ethanol is predominantly used in blending with gasoline for direct use in conventional internal combustion engines. While blending limits vary depending on location and use, limits have historically been put in place in part to protect older engines, many of which were not designed to run on higher ethanol concentrations.^{3,4} In addition to blend limitations, ethanol has only two thirds the energy density of gasoline and is fully miscible with water. Higher alcohols, particularly *n*-butanol, could offer some advantages in gasoline blending due to their higher energy contents and immiscibility with water.⁵ Additionally, while ethanol is industrially produced from fermentation of biomass-derived sugars, *n*-butanol is obtained from non-renewable feedstocks, specifically *via* hydrogenation of butyr-

aldehyde produced *via* hydroformylation of propylene.^{6–8} *N*-Butanol possesses a variety of other commercial applications as a solvent, perfume additive and flavoring agent.⁷

An alternative to petrochemical *n*-butanol production involves the alcohol coupling process known as Guerbet condensation. This process, as first described by Marcel Guerbet in 1899, involves three key steps: dehydrogenation of an alcohol to an aldehyde, aldol condensation between aldehydes to an unsaturated aldehyde, and full hydrogenation to a higher alcohol.⁹ Recently, interest in this area has been revived by a desire to valorize lighter alcohols such as ethanol.^{10–12} This chemistry has been accomplished with a range of heterogeneous catalysts, which possess both acid and base sites. In the absence of a transition-metal promotor, however, catalysts typically do not possess sufficiently high activity at temperatures below 350 °C, above which alcohol selectivity decreases as side-reactions such as ethanol dehydration become more prominent.^{13–15} Aldol condensation can be effectively carried out with these catalysts at lower temperatures, though alcohol conversion can only be achieved if dehydrogenation can be facilitated.^{16,17} This reaction can be accelerated by the addition of transition metals such as Cu or Ni which facilitate H-transfer to accelerate alcohol–aldehyde interconversion, which otherwise is believed to require intermolecular hydrogen transfer.^{18–21} Cu sites have additionally been shown to catalyze both aldolization and esterification.^{22–25}

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Mixed Mg–Al oxides have received considerable attention for the Guerbet condensation of ethanol.^{10,14,15,26} These Mg–Al mixed oxides are conventionally prepared *via* decomposition of hydrotalcites which are synthesized *via* coprecipitation of Mg and Al precursors (*e.g.* nitrates) at high pH. In our previous work, we have used an emulsion-mediated synthesis method to produce these materials with additional porosity, where the formation of a sol–gel transition in an emulsion acts as a soft template with a nonpolar phase dispersed in an aqueous phase.^{27–29} Inorganic precursors organize around the droplets of the nonpolar phase, which then chemically gel the sol. The removal of droplets upon drying and calcination creates a unique pore structure. Increasing the surface area of a material increases the potential number of available catalytically active sites and promotes the dispersion of added metals.^{30–34} Furthermore, the presence of amphipathic molecules and apolar molecules in the synthesis can alter the strength of the sites in the catalyst due to interactions between the templates and the surface of the metal oxides.^{35,36} In this way, the efficiency of the material can be increased by modifying its structural and chemical properties, the composition, and organization of the surface atoms, as shown previously.^{27,35,37,38} We have specifically shown that the catalytic activity of alumina and AlMgO for ethanol dehydration and 2-propanol conversion, respectively, strongly depends on the pore structure produced with this method.^{29,36} We additionally produced efficient ethanol dehydrogenation catalysts *via* the addition of Cu to porous alumina catalysts.^{22,23,37}

In this work, we evaluate the effect of textural properties and acid–base properties of AlMgO and AlCaO oxides with and without the addition of Cu for the condensation of ethanol at elevated hydrogen pressures. Our previous studies of ethanol conversion over Cu-promoted alumina did not show extensive C–C coupling, which is typically slow for alumina compared to Mg–Al oxides.²⁶ The substitution of magnesium by calcium is intended to change the basicity and acidity of the material and thus influence catalytic performance.³⁸ Here high hydrogen pressures (32 bar) are utilized to promote the saturation of Guerbet intermediates and increase the selectivity of aldol condensation products.

2. Experimental

2.1. Catalyst preparation

The synthesis of the porous oxides involved the use of aluminum tri-*sec*-butoxide (Al(O-*s*-But)₃) and magnesium nitrate or calcium nitrate. The assembly of the emulsified system occurred by the simultaneous utilization of *n*-dodecane, Pluronic P123 (block copolymer, 20 EO : 70 PO : 20 EO, Mw. of 5826 g mol^{−1}) and ethanol as solvent, where Pluronic P123 and *n*-dodecane were the organic pore structure directing agents (PSDA).^{29,35}

The dispersion of the *n*-dodecane in ethanol occurred at room temperature. The final mole composition of the sol was: 0.01 Pluronic: 1 (Al(O-*s*-But)₃): 1 Mg(NO₃)₂ or Ca(NO₃)₂:

15 ethanol. A 29 wt% solution of ammonium hydroxide was added dropwise under stirring to produce the sol–gel transition (pH 10). The reference samples were prepared in the same procedure, though without addition of PSDA. The as-synthesized samples were calcined at 700 °C for 6 h under air flow to produce mixed oxides. For synthesis of Cu-doped catalysts, the mixed oxides were then dispersed in Milli-Q water, and tetraamminecopper(II) sulfate monohydrate [Cu(NH₃)₄SO₄·H₂O, Sigma-Aldrich] was added to the suspension under continuous mixing at room temperature. The pH was adjusted to 9 through dropwise addition of dilute 2 mol L^{−1} nitric acid (Sigma-Aldrich). The mixture was aged for 20 h while being stirred at room temperature, and the solid material was then filtered and washed with deionized water. The sample was dried overnight at 110 °C in air. The dried catalyst was pretreated in a flow-through cell at a temperature of 500 °C (with a heating rate of 1 °C min^{−1}) under a helium flow (30 mL min^{−1}) for 3 h, followed by air flow, switched to He for 20 min, reduced under hydrogen flow, and finally passivated at room temperature with 1% O₂ in He. AlMgO and AlCaO catalysts were prepared with and without an emulsion, with the former denoted by “-P” to imply the added porosity. All catalysts were examined for catalytic activity with and without the addition of Cu.

2.2. Methods for the characterization of structure of oxides

Nitrogen adsorption–desorption isotherms were recorded at −195 °C and *P*/*P*₀ ranging between 0.001 and 0.998, using a Micromeritics ASAP 2020 instrument. Samples were degassed prior to analysis at 200 °C for 6 h under a vacuum of 10 μPa. The specific surface area was calculated according to the BET equation³⁹ and the pore size distribution was determined using the BJH equation.

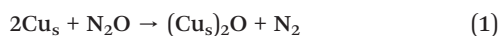
The thermal behavior of the AlMgO and AlCaO samples, calcined and exposed to atmospheric environment, was investigated by thermogravimetric analysis, performed from room temperature up to 900 °C under air flow (50 mL min^{−1}), by using a SDT600 TA Instruments at a heating rate of 10 °C min^{−1}.

Scanning transmission electron microscopy (STEM) was performed using a FEI Titan STEM instrument with a Cs probe aberration corrector operated at 200 kV with a spatial resolution of <0.1 nm. A high-angle annular dark-field (HAADF) mode with a HAADF detector angle ranging from 54 to 270 mrad, a probe convergence angle of 24.5 mrad, and a probe current of approximately 25 pA was used to record the images. Catalyst samples were suspended in ethanol and then dropped on a holey carbon Au TEM grid. Samples were plasma cleaned before being loaded into the microscope. Particle sizes were calculated based on a minimum of 800 particles for each sample.

The Cu loading of catalysts were determined with inductively coupled plasma absorption emission spectroscopy (ICP-AES) using a PerkinElmer Plasma 400 ICP emission spectrometer. Typically, 20 mg catalyst samples were digested with a

mixture of 3 mL of nitric acid (Fischer, 65%) and 7 mL of hydrochloric acid (Fischer, 37%) in a Teflon beaker at 120 °C. The digested mixture was cooled to room temperature, diluted in water, filtered, and analyzed. Cu standards for ICP analyses were prepared from a concentrated Cu ICP standard (Fluka, $1000 \pm 2 \text{ mg L}^{-1}$).

The number of Cu surface sites was determined using reactive N_2O chemisorption.⁴⁰ Prior to the measurement, catalysts were reduced *in situ* at 500 °C (with a heating rate of $1 \text{ }^\circ\text{C min}^{-1}$) under a H_2 flow for 2 h. After reduction, the samples were cooled to 90 °C, and the cell was evacuated to 10^{-5} Torr. N_2O was then introduced and reacted with surface metallic Cu to produce N_2 gas and chemisorbed O. The stoichiometry for the adsorption of O on the Cu surface was assumed to be 1/2 based on eqn (1) where Cu_s denotes as Cu surface sites.



The amount of O adsorbed on the surface was quantified by monitoring the N_2 pressure using a gas handling system and a volumetric system employing Baratron capacitance manometers for precision pressure measurement (5×10^{-5} Torr) after condensation of N_2O in a cold trap using liquid nitrogen.

Acidities and basicities of the samples were determined by temperature programmed desorption of ammonia (NH_3 -TPD) and carbon dioxide (CO_2 -TPD), respectively, using a Micromeritics AutoChem II 2920 apparatus. Samples were pretreated in H_2 (30 mL min^{-1}) at 500 °C ($1 \text{ }^\circ\text{C min}^{-1}$) and held for 2 h to remove adsorbed species and to reduce the Cu for consistency with catalytic experiments. Prior to NH_3 -TPD, samples were then cooled to 150 °C in He (50 mL min^{-1}) prior to NH_3 adsorption. A stream containing 1 vol% NH_3 in He (50 mL min^{-1}) was introduced for 30 min until adsorption equilibrium was achieved. Physisorbed NH_3 was then purged at 150 °C under a flow of He for 60 min. Chemisorbed NH_3 was then desorbed in a 50 mL min^{-1} flow of He heating from 150 to 500 °C at a rate of $10 \text{ }^\circ\text{C min}^{-1}$ with the NH_3 evolution monitored by an online mass spectrometer (Cirrus 2, MKS Instruments), which was integrated to determine the number of acid sites and deconvoluted to determine acid strengths. CO_2 -TPD was performed in the same manner but with the uptake conducted at 50 °C in a stream of 50 vol% CO_2 in He (30 mL min^{-1}) for 1 h followed by 1 h of He flow to remove weakly adsorbed CO_2 . The temperature was then increased to 500 °C at a rate of $10 \text{ }^\circ\text{C min}^{-1}$ in He (30 mL min^{-1}) with CO_2 evolution monitored by an online mass spectrometer.

The surface acid sites of Cu samples were further analyzed by Fourier transform infrared spectroscopy (FTIR) using a Shimadzu Prestige 21 spectrophotometer with pyridine as the probe molecule. The spectral resolution was 4 cm^{-1} , and 40 scans were accumulated. For each sample, 15 mg of catalyst were pressed into 13 mm diameter pellets without dilution. All catalysts were reduced under 25 mL min^{-1} 4% H_2/N_2 at

$1 \text{ }^\circ\text{C min}^{-1}$ to 500 °C with a 2 h hold. After cooling to 100 °C, N_2 was flowed for 1 h to remove physisorbed H_2 . After cooling to room temperature, N_2 was flowed through a pyridine bubbler and over the catalyst for 5 min, followed by a switch to N_2 without pyridine. The catalysts were heated at $1 \text{ }^\circ\text{C min}^{-1}$ to 150 °C and held for 1 h to remove physisorbed pyridine. Samples were then cooled to room temperature prior to data collection. Spectra were recorded at room temperature with background subtraction based on a previously-recorded spectrum of the dehydrated sample. The relative contributions of Lewis and Brønsted acid sites after adsorption and partial desorption of pyridine were obtained by integration of the bands at 1440 cm^{-1} (Lewis sites) and 1540 cm^{-1} (Brønsted sites). The relative concentration Brønsted/Lewis acid sites was determined by means of the Beer-Lambert Law for an Emeis molar extinction factor of 1.67 cm mol^{-1} for band at 1540 cm^{-1} and 2.22 cm mol^{-1} for band at 1450 cm^{-1} .

2.3. Catalytic activity

Ethanol coupling reactions were performed in fixed-bed stainless-steel tubular flow reactors (40 cm long, 0.46 cm inner diameter) heated with a tube furnace (Lindberg Blue M Mini-Mite) with a 30 cm heated region. Reactors were packed with 225–675 mg of catalyst at the center of the heated region with quartz beads and quartz wool on either side. An aluminum block was utilized between the furnace coils and reactor to maintain a 30 cm isothermal region. Temperature was measured *via* a thermocouple positioned alongside the reactor with the tip at the center of the bed. Prior to reaction, all samples were pretreated in H_2 (100 mL min^{-1}) at 500 °C ($1 \text{ }^\circ\text{C min}^{-1}$, 2 h) with the flow rate controlled *via* a mass-flow controller (SLA 5850, Brooks Instruments). During reaction, neat ethanol (Decon's Pure Ethanol, 200 proof) was fed at flow rates between 10 and $30 \text{ } \mu\text{L min}^{-1}$ *via* a syringe pump (Teledyne Isco 500D) into a stream of 30–90 standard mL min^{-1} of H_2 flowing downward through a preheated region prior to the reactor. All reactions were performed at 325 °C. Below the furnace products were fed through a 300 mL condenser kept at 0 °C *via* an ice bath. Species which did not fully condense then passed through a back-pressure regulator set at 32 bar to an online GC (Shimadzu GC-2014) equipped with both a flame-ionization detector (FID) and a thermal conductivity detector (TCD). The gas was analyzed every 4–12 h directly prior to liquid collection. The condenser contents were collected directly into 1,4-dioxane to fully solubilize all components and minimize flashing of volatile components. Samples were then further diluted in a 1,4-dioxane solution containing a *n*-decane tracer (1.5 wt%) and analyzed *via* GC FID (Shimadzu GC-2010). Product identification was validated using GC MS for selected samples. Quantification was performed using response factors developed from external calibration curves. Compounds for which standards could not be easily obtained were quantified using their effective carbon numbers according to Scanlon and Willis,⁴¹ though

all major components were quantified using standards. Reactions were performed for 32–80 h. Carbon balances (carbon detected in the product minus the carbon fed into the reactor) for reported data were between 95 and 105%. Tabulated values are averages based on time points which demonstrate carbon balances in this regime, typically three points. Yields, conversion, and selectivities are defined on a carbon basis (referred to as C%). Conversion is calculated as the sum of yields rather than directly from the change in ethanol flow rate in order to minimize amplification of minute analytical errors. This includes an approximate yield to species detected *via* GC that were not identifiable.

3. Results and discussion

3.1. Structural characteristics of the catalysts

Textural properties of the mesopores were determined by nitrogen physisorption measurements and the BJH pore size distribution curves obtained from the desorption branches of the isotherms (Fig. 1). The emulsion method proved to be effective for the creation of more porous materials (Table 1), when comparing with the reference samples (AlMgO and AlCaO). The hysteresis between the adsorption and desorption curves is typical evidence of the existence of mesopores, shown in AlMgO-P and AlCaO-P. However, the absence of a plateau in the P/P_0 region near 1 is characteristic of the type II isotherms often observed for solids in which macropores are present as well. According to the IUPAC classification, the shallow slopes of the isotherms for the samples, together with the non-parallel behavior of the adsorption/desorption curves, indicate a H3 hysteresis cycle.

Physisorption measurements were additionally performed on catalysts with copper. The shape of the hysteresis does not change significantly, however the pore diameter greatly decreases. The addition of copper likely blocks certain pores of the material which also leads to lower BET surface areas. The pore distribution also shows a well-defined porous struc-

ture for the porous catalyst with diameters of 10 and 15 nm for Cu/AlMgO-P and Cu/AlCaO-P, respectively, according to Fig. 1b and d. The emulsion method also increased BET surface area relative to the reference catalysts.

Cu particles were examined by STEM. Fig. 2 shows representative transmission electron micrographs and the corresponding particle size distributions for Cu catalysts after reduction and passivation. Cu/AlMgO-P exhibits a lower average Cu particle size and more narrow distribution than the reference material, indicating a more homogeneous surface for the porous support. The average size of the Cu particles decreased from 3.7 to 2.1 nm when using the porous support. This behavior was not observed for the Ca samples, however, as the emulsion-mediated method resulted in a slight increase in Cu particle size from 2.6 to 3.1 nm without a large effect on the sharpness of the distribution.

Cu site densities were additionally quantified using reactive N_2O chemisorption, which increased for both AlMgO and AlCaO when an emulsion-mediated synthesis was used. The AlCaO catalysts generally possessed higher Cu site densities, which is in part due to their higher Cu loadings.

The acid and base properties of the catalysts were examined *via* NH_3 -TPD and CO_2 -TPD, respectively, with the smoothed profiles shown in Fig. S1.† Peak deconvolution with Gaussian profiles was performed to separate the CO_2 -TPD profiles into weak, medium and strong sites and the NH_3 -TPD profiles into weak and strong sites (Tables 1 and S1†), based on the methods used by Di Cosimo *et al.* with Mg_xAlO_x catalysts.^{25,26} The emulsion method was found to increase both the acid and base site densities for AlMgO as a consequence of an increase in surface area. AlCaO catalysts were found to possess reduced acidity and basicity compared to AlMgO with less than 25% of the base sites of AlMgO catalysts and less than 6% of the acid sites, which were too low for quantification with AlCaO. The lower basicity of AlCaO can be attributed to surface carbonates, which remain present even after 500 °C calcination as shown by

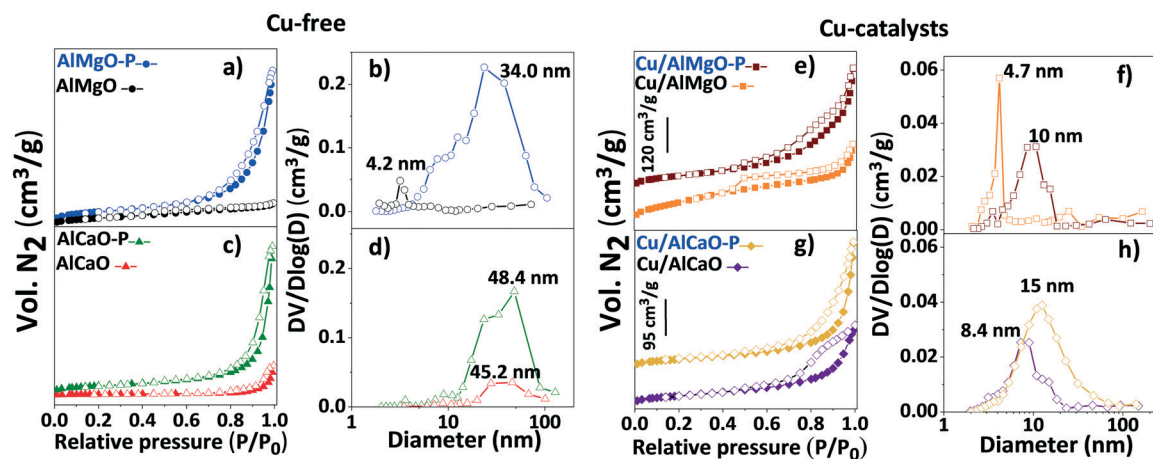


Fig. 1 (a, c, e and g) Nitrogen adsorption–desorption isotherms and (b, d, f and h) BJH pore size distributions of the Cu-free and Cu-catalysts on AlMgO and AlCaO.

Table 1 BET area, Cu loading, Cu particle size, Cu⁰ site density, and distribution of strengths of surface acid and basic sites

Sample	BET area (m ² g ⁻¹)	Cu loading ^a (%)	Cu particle size ^b (nm)	Cu ⁰ sites ^c (μmol g ⁻¹)	Total NH ₃ ^d (μmol g ⁻¹)	Total CO ₂ ^e (μmol g ⁻¹)
Cu/AlMgO	106	8.0	3.7	67	92.9	115
Cu/AlMgO-P	171	8.0	2.1	91	106	78.6
Cu/AlCaO	102	9.0	2.6	118	10.1	52.4
Cu/AlCaO-P	120	9.6	3.1	140	12.5	49.7
AlMgO	191	—	—	—	76.9	192
AlMgO-P	315	—	—	—	85.6	238
AlCaO	133	—	—	—	— ^f	43.8
AlCaO-P	202	—	—	—	4.0	46.9

^a Cu loading determined by ICP. ^b Cu particle sizes obtained from STEM. ^c Metallic copper sites quantified by N₂O chemisorption. ^d Acid sites quantified from NH₃-TPD profiles. ^e Base sites quantified with CO₂-TPD profiles. ^f Without significant acidity.

thermogravimetric analyses (Fig. S2†). After calcination and exposure to atmosphere, AlCaO-P continues to lose appreciable weight above 600 °C while minimal weight loss occurs for AlMgO-P above 500 °C. For all Cu-free catalysts, both acid and base strengths were fairly evenly distributed according to the temperature-based deconvolution method utilized. Upon addition of Cu, for all catalysts the number of acid sites increased with a shift toward stronger acidity while the Cu/AlMgO catalysts remained more acidic than the Ca counterparts. As Cu is expected to block surface sites, this would suggest that the acidity may come from Lewis-acidic oxidized Cu in the framework.⁴² A slight shift toward stronger sites is also observed. The general trends between porous and reference samples are maintained. Cu addition greatly decreases the number of base sites available on AlMgO catalysts, while

AlCaO catalysts experience a more minor increase. In all cases, Cu addition decreases the average base site strength. The high sensitivity of base site density to Cu addition with AlMgO *versus* AlCaO is unclear, though it may be related to preferential binding of Cu to certain base sites which are uniquely present on the former.

The distributions of Brønsted *versus* Lewis acid sites on Cu catalysts were determined *via* pyridine FTIR analyses in the range of 1350–1650 cm⁻¹ (Fig. 3). The absence of a band near 1540 cm⁻¹ indicative of pyridinium cations (PyH⁺) reflects the lack of Brønsted acid sites (B), while a band centered at 1450 cm⁻¹ – corresponding to pyridine chemisorbed on Lewis acid sites (L) – was clearly observed. The ratio of Brønsted sites to total acid sites (*T*) was found to be close to zero for all catalysts, indicating that only Lewis acidity is present. A physisorbed pyridine band is observed for AlCaO catalysts, which is a result of the overall low acidity of these catalysts as supported by NH₃-TPD measurements. These results are in agreement with those of Lercher and Prescott *et al.*^{43,44} who observed only Lewis acidity on Al₂O₃-MgO catalysts.

3.2. Catalytic performance in high-pressure Guerbet coupling of ethanol

Ethanol conversion over mixed-metal oxides doped with transition metals produces various species through a variety of

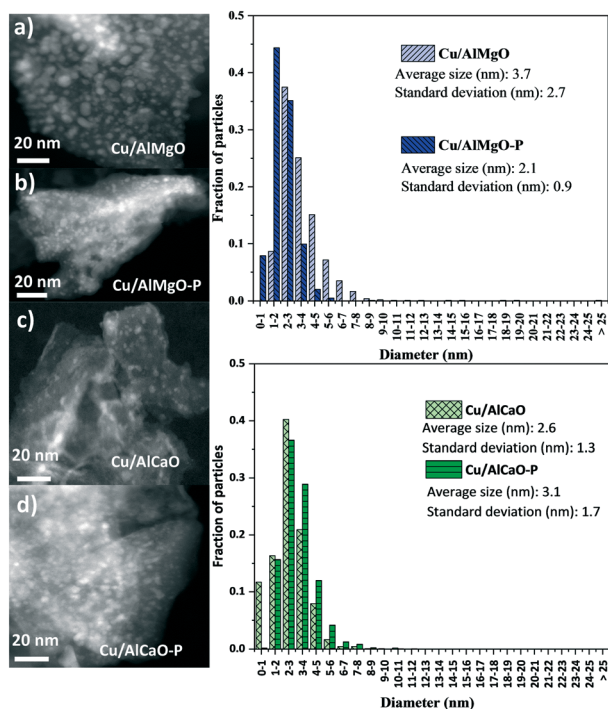


Fig. 2 Transmission electron microscopy images and particle size distributions.

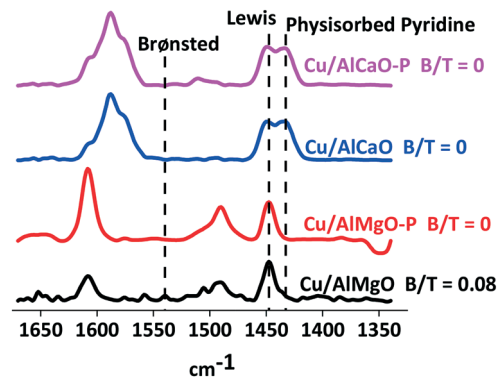


Fig. 3 IR spectra of adsorbed pyridine on Cu catalysts.

pathways as generalized in Fig. 4. The condensation of ethanol (EtOH) produces higher alcohols and aldehydes, while side reactions are commonly present, which include intramolecular dehydration to olefins, intermolecular dehydration to ethers, isomerization of intermediates to produce ketones and secondary alcohols and esterification. Alcohol condensation occurs either through formal Guerbet coupling, in which carbon-carbon bonds are formed explicitly through aldehyde aldolization, or through direct condensation mechanisms, in which the reactant alcohols are directly condensed. On mixed oxides such as AlMg_xO_y and hydroxyapatite, which do not possess transition metals capable of dissociating H_2 , the hydrogen transfers involved in formal Guerbet condensation mechanisms cannot be facilitated through gas phase H_2 .^{20,21,45–47} As a result, with these catalysts hydrogen transfers occur predominantly *via* intermolecular transfer hydrogenations. The addition of Cu allows for H_2 activation, providing more control over the ethanol dehydrogenation equilibrium, which strongly depends on the reaction pressure and the partial pressure of H_2 in the reactor feed.¹⁸

The effect of varying system pressure on ethanol coupling over Cu/AlMgO in a H_2 atmosphere (12% ethanol) was examined between 1 and 31 bar total pressure at 325 °C as shown in Fig. 5. At atmospheric pressure the catalyst produced predominantly acetaldehyde at 17.9% conversion with a selectivity of 56%, while the desired coupling products (C_{4+} alcohols and aldehydes) were produced with only 22.1% selectivity. Increasing the pressure to 10 bar doubled the selectivity to desired products (to 43.1%) while the acetaldehyde selectivity was halved (to 26.7%) and the conversion was increased slightly (to 21.1%). Further increasing the pressure to 24 and then 31 bar had a negligible impact on conversion (to 19.5

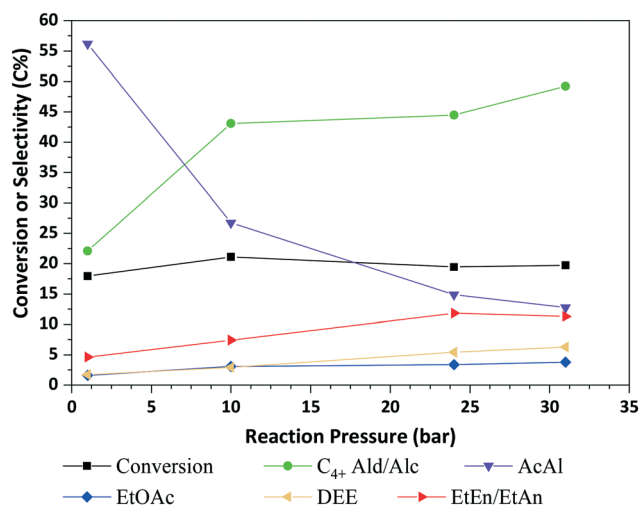


Fig. 5 Ethanol conversion over Cu/AlMgO at total pressures of 1, 10, 24, and 31 bar. Conditions: 325 °C, 225 mg catalyst, $10 \mu\text{L min}^{-1}$ ethanol, $30 \text{ mL min}^{-1} \text{H}_2$. C_{4+} Ald/Alc = C_{4+} aldehydes and alcohols (desired products), AcAl = acetaldehyde, EtOAc = ethyl acetate, DEE = diethyl ether, EtEn/EtAn = ethylene and ethane.

and 19.7%, respectively) while the desired product selectivity increased (to 44.5 and 49.2%, respectively) and the selectivity to acetaldehyde decreased (to 14.9 and 12.8%, respectively). The fairly constant conversion observed over the entire pressure range is due to the simultaneous decrease in the yield of acetaldehyde (10.1 to 2.5%) and increase in the yield of all other products (7.9 to 17.2%). The former is related to ethanol dehydrogenation equilibrium, as will be discussed in further detail, while the latter is related to the increased catalyst surface coverages of reactants at elevated pressures.

Ethanol-acetaldehyde equilibrium can be modeled using tabulated thermodynamic data to provide further insight into the effects of changing the reaction pressure (Table S2†).⁴⁸ With a 12% ethanol/ H_2 feed at 325 °C, the equilibrium molar ratio of acetaldehyde to ethanol decreases (1.22, 0.11, 0.06, 0.04) as the system pressure increases (1, 20, 24, 31 bar). At atmospheric pressure, the experimental ratio is far below equilibrium (0.12 *versus* 1.22), though the equilibrium ratio is approached at higher pressures (0.03 *versus* 0.04 at 31 bar). While the mole fraction of acetaldehyde therefore decreases as pressure increases, the partial pressure of acetaldehyde still increases since the total pressure of the system does as well. The equilibrium pressure of acetaldehyde therefore increases over this range (6.4, 13.4, 14.5, 14.7 kPa), which asymptotically approaches ~15 kPa upon further increases in pressure. Reaction rates which depend on acetaldehyde partial pressures, such as aldolization, are therefore expected to improve with increases in the reaction pressure, especially between 1 and 15 bar. Catalyst surface coverages are also expected to follow an asymptotic trend as full saturation is approached, however, thus the effect of pressure is too complex to establish a clear causal link between acetaldehyde pressure and condensation rates. The equilibrium pressure of acetaldehyde can be further increased by operating with

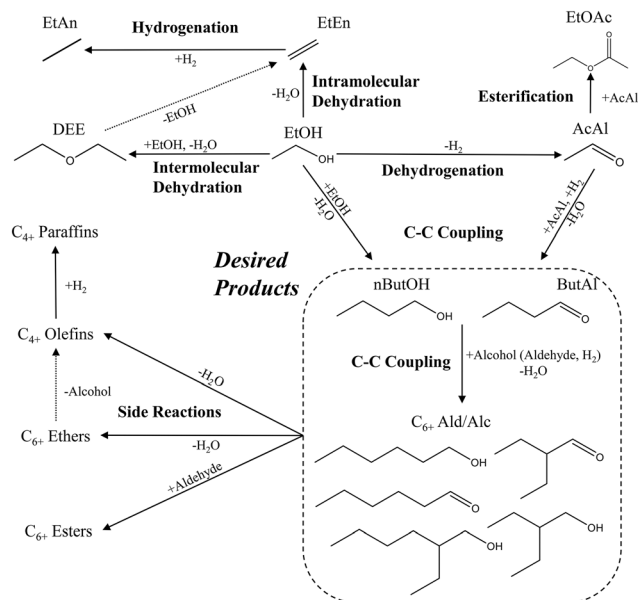


Fig. 4 Major products and generalized reaction pathways involved in the high-pressure conversion of ethanol over mixed-metal oxides doped with transition metals.

an inert gas co-feed. Operating at 31 bar with Ar instead of H_2 increases the equilibrium acetaldehyde pressure by over an order of magnitude from 14.7 to 151.7 kPa while the ratio of acetaldehyde:ethanol increases from 0.04 to 0.72. Experimentally, operating with an Ar feed produced an acetaldehyde:ethanol ratio of 0.07, far below equilibrium. This was accompanied by a large increase in conversion from 19.7 to 32.5%. The main product was ethyl acetate (34.6% selectivity), which is produced at much lower levels with a H_2 atmosphere (0.7 versus 11.2% yield). Simultaneously, the yield of desired products decreased by about 40% (9.7 to 5.7%). This high esterification selectivity has also been noted previously in batch reactions of Cu-doped $AlMg_xO_y$ in air.⁴⁹ High H_2 pressures therefore inhibit esterification, which is known to be catalyzed by Cu sites *via* the Tischenko reaction.^{25,50} Whether this is caused by alteration of the Cu oxidation state or competitive adsorption between alcohols and aldehydes with H_2 is uncertain. Altogether, these results provide a general understanding of how the gaseous environment can be manipulated to promote Guerbet coupling when using catalysts capable of mediating ethanol dehydrogenation through H_2 . Increasing pressure promotes condensation while favourably decreasing the acetaldehyde selectivity. These effects are strongest up to ~ 15 bar, above which increasing pressure has a minimal impact, in qualitative agreement with thermodynamic calculations. Such calculations are expected to be particularly useful in guiding the use of catalysts, which are more active for dehydrogenation than the Cu/AlMgO-P used here, since the activity was not sufficient to achieve equilibrium under most conditions.

Based on these findings, the catalysts synthesized in this study were compared in a H_2 environment at 32 bar total pressure. Stable performance was observed for all catalysts between 10 and 30 h time-on-stream (TOS). Conversions and selectivities for Cu-doped catalysts are reported in Fig. 6.

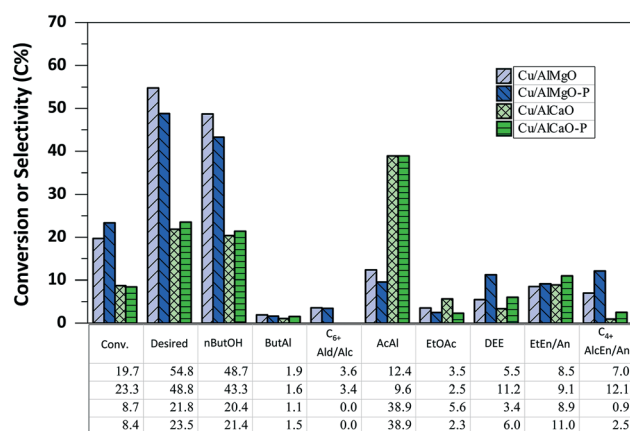


Fig. 6 Conversion and selectivities to major species in ethanol conversion over Cu-doped catalysts. Conditions: 325 °C, 225 mg catalyst, 10 $\mu L \min^{-1}$ ethanol, 30 $mL \min^{-1}$ H_2 , 32 bar total pressure. nButOH = n-butanol, ButAl = butyraldehyde, Ald/Alc = aldehydes and alcohols, AcAl = acetaldehyde, EtOAc = ethyl acetate, DEE = diethyl ether, EtEn/An = ethylene and ethane, AlcEn/An = alkenes and alkanes.

Cu/AlMgO and Cu/AlMgO-P showed similar activities with conversions of 19.7 and 23.3%, respectively. These catalysts were more active than their Ca counterparts with conversions of 8.7 and 8.4% for Cu/AlCaO for Cu/AlCaO-P, respectively. At these conditions, Cu/AlMgO and Cu/AlMgO-P were also more selective to desired condensation products (54.8 and 48.8%, respectively) than Cu/AlCaO and Cu/AlCaO-P (21.8 and 23.5%, respectively). The Ca catalysts were more selective to acetaldehyde at these conditions, though the acetaldehyde yield was similar for all catalysts. As noted earlier, acetaldehyde is produced at nearly equilibrium levels on these catalysts, thus it may be more informative to compare product selectivities with the omission of acetaldehyde. In this case, Mg catalysts remain more selective to C_{4+} aldehydes and alcohols than the Ca catalysts with adjusted selectivities of 62.5% and 54.0% for Cu/AlMgO and Cu/AlMgO-P, respectively, versus 35.7% and 38.5% for Cu/AlCaO and Cu/AlCaO-P, respectively. These comparisons are notably made at different conversions, though it is unlikely that Ca catalysts will show improved adjusted selectivities upon increasing conversion.

In the absence of Cu, lower activities were observed with AlMgO and AlMgO-P achieving only 7.1 and 11.1% conversion, respectively and AlCaO and AlCaO-P demonstrating essentially no activity with $\sim 0.2\%$ conversion (Fig. S3†). The particularly low activities of the Cu-free Ca catalysts are consistent with their low numbers of acid and base sites compared to the Mg catalysts, as measured by TPD. Upon Cu addition, the acetaldehyde yield increased by 10–15 times for the Mg catalysts and 70–75 times for the Ca catalysts, consistent with the observation by Gines *et al.* that ethanol dehydrogenation rate over K-Mg₅CeO_x at atmospheric pressure increased by nearly two orders of magnitude upon addition of 7 wt% Cu, allowing for equilibration to occur at longer contact times.¹⁸ The desired product yield also increased by 2–5 times for the Mg catalysts and 14–30 times for the Ca catalysts upon Cu addition. These increases may be due to either (1) the increase in acetaldehyde pressures resulting from Cu-catalyzed ethanol dehydrogenation, (2) the addition of sites which enable H-transfer between the support and the gas-phase, or (3) the addition of Cu sites capable of catalyzing condensation reactions independently from the support. The first would be true if aldolization is a key condensation pathway. The second has notably been proposed by Gines *et al.* to explain the improved condensation rates with Cu addition to K-Mg₅CeO_x at atmospheric pressure with an inert co-feed gas.¹⁸ The third has been noted to occur over Cu, though it is typically observed alongside significant esterification.²⁵ The role that Cu plays is dependent on the mechanism of alcohol condensation, which has been debated in the literature. The two general mechanisms proposed for alcohol condensation on heterogeneous catalysts are the Guerbet coupling reaction, which formally involves aldolization with aldehyde intermediates, and direct alcohol condensation, wherein aldehydes do not serve as key intermediates. The relevance of aldolization has been challenged by Meunier *et al.* who showed

that a formal aldol condensation mechanism was not thermodynamically consistent with the ratio of condensation products *versus* acetaldehyde observed in reactions at atmospheric pressure with a hydroxyapatite catalyst.⁵¹ Evidence of both condensation mechanisms occurring simultaneously has also been found in plots of the *n*-butanol formation rate *versus* contact time both on hydroxyapatite and AlMg₂O₃, which show inflections that suggest *n*-butanol is formed both as a primary product (direct ethanol condensation) and as a secondary product (acetaldehyde aldolization).^{45,52} An aldehyde-mediated route must play some role for the Cu/AlMgO and Cu/AlCaO catalysts based on the butyraldehyde:*n*-butanol ratios observed. If only direct condensation mechanisms are present, butyraldehyde can only be produced *via* *n*-butanol dehydrogenation, thus the butyraldehyde:*n*-butanol ratio must be less than or equal to the equilibrium ratio (estimated to be 0.034 at these conditions). The calculated ratio is higher than this for all catalysts (between 0.037 and 0.071), however, thus an alternate condensation mechanism must be present to explain butyraldehyde formation. Additionally, the improved condensation yields measured upon Cu addition are more mechanistically consistent with promotion of an aldehyde-mediated route than a direct condensation route. The possibility that both direct and indirect condensation mechanisms are active cannot be discounted, however, especially for Cu/AlMgO catalysts. Given that AlCaO and AlCaO-P are essentially inactive before Cu addition, the indirect route is also most likely dominant for Cu/AlCaO catalysts.

Other than acetaldehyde, the main side-products observed with Cu-doped catalysts were ethylene/ethane (EtEn/An, 8.5–11.0% selectivity), C₄₊ alkenes/alkanes (C₄₊ AlkEn/An, 0.9–12.1%), diethyl ether (DEE, 3.4–11.2%), and ethyl acetate (EtOAc, 2.3–5.6%). In the absence of Cu, diethyl ether is the dominant side-product on Mg catalysts with ~50% selectivity. The higher yield of diethyl ether *versus* ethylene has been commonly observed over AlMg₂O₃ catalysts.^{26,53} Diethyl ether production on Mg catalysts is greatly inhibited upon Cu addition, thus dispersed Cu particles may cover sites which promote intramolecular dehydration. Another possibility is that new sites are formed upon Cu addition which promote dehydration of diethyl ether to ethylene, since the yield to ethylene and ethane increase upon Cu addition (with the latter forming *via* ethylene hydrogenation). Both possibilities are consistent with the simultaneous decrease in basicity (thought to catalyze intramolecular dehydration) and increase in acidity (thought to catalyze intramolecular dehydration) upon Cu addition to Mg catalysts, as measured by TPD.²⁶ Ethyl acetate is also absent on Cu-free catalysts, consistent with the aforementioned expectation that esterification is promoted *via* Cu sites as is known in the literature.^{25,50}

Previous studies on ethanol to ethene, diethyl ether, acetaldehyde and/or ethyl acetate pointed out that the support porosity played an important role to give the highest selectivity to the targeted product.^{22,23,35} Therein the synthesis

method was important because it influenced acidity/basicity, and the formed porosity influenced copper particle size and the oxidation state of copper, properties that were fine-tuned on a porous support. Application of the emulsion-mediated method altered the textural properties of the catalysts and improved the overall catalytic performance. However, these changes did not reflect improvement in the Guerbet reaction as occurred before, as it was mostly influenced by reaction pressure and the presence of copper. AlCaO catalysts were employed to increase basicity and decrease acidity relative to AlMgO in order to promote condensation reactions while reducing ethanol dehydration. Though dehydration decreased, *n*-butanol yield also decreased possibly because CaO is prone to carbonation in ambient air. These carbonates are highly stable and require high temperatures to decompose. For instance, Granados *et al.* found that amorphous carbonates formed on CaO due to air exposure could not be removed until above 700 °C are often required to remove adsorbed CO₂ and moisture to produce an active catalyst.^{54,55} These catalysts were therefore determined to be generally inferior to their Mg counterparts.

To achieve higher conversions, ethanol coupling over Cu/AlMgO-P was examined as a function of WHSV with a single catalyst bed (675 mg) as shown in Fig. 7. Initially a WHSV of 2.10 h⁻¹ was used for a direct comparison with the previous studies using lower catalyst loadings (225 mg) in order to test for transport limitations. The higher catalyst loading lead to a negligible increase in conversion (22.9 to 23.3%) while the selectivity to *n*-butanol decreased (43.3 to 35.8%) and the diethyl ether selectivity increased (11.2 to 19.7%). These selectivity shifts are more likely to result from catalyst batch deviations than transport limitations. In addition to these experimental results, calculations provided in the ESI† suggest that such limitations are not present. Decreasing the WHSV to 1.40 h⁻¹ and then to 0.70 h⁻¹ increased conversion

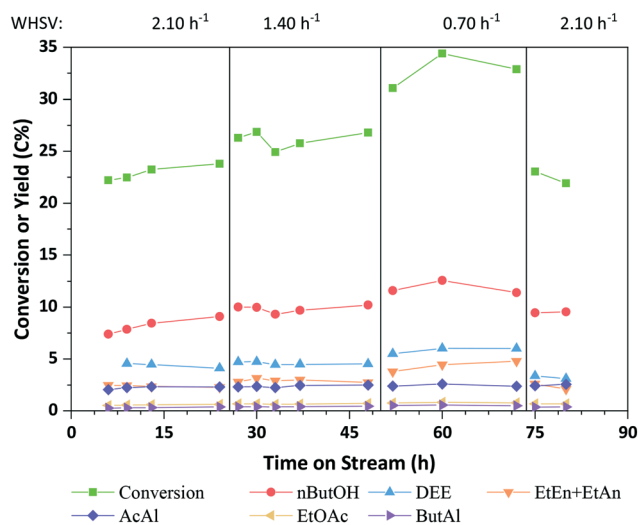


Fig. 7 Conversion and yields in the conversion of ethanol over Cu/AlMgO-P at varying WHSV. Conditions: 325 °C, 675 mg catalyst, 10–30 μL min⁻¹ ethanol, 30–90 mL min⁻¹ H₂, 32 bar total pressure.

to 26.1 and 32.8%, respectively. These increases are lower than expected assuming that ethanol conversion can be approximated using a power law rate expression with an ethanol order of two or lower. First, second, and even third order rate expressions would predict ethanol conversion to increase by 140%, 110% and 90%, respectively, when decreasing the WHSV from 2.10 to 0.70 h⁻¹, though the conversion increased by only 40%. This therefore suggests either severe irreversible deactivation or reversible inhibition by strongly adsorbing products. Irreversible deactivation is not responsible for these trends since returning to the original WHSV showed that the catalytic activity had not appreciably changed over 75 h. These trends are more therefore more likely due to reversible blocking of sites by species produced during the reaction, limiting ethanol adsorption either *via* competitive adsorption or reversible site alteration. Previous researchers have shown that water inhibits ethanol coupling over decomposed Cu_xMg_yAlO_z hydrotalcites in batch mode, presumably *via* the conversion of Lewis basic O²⁻ sites into weaker Brønsted OH⁻ sites.⁵⁶ Minimal deviations in product distribution were observed over this WHSV range, which suggested that all reactions are inhibited to similar extents. Heavier condensation products (*i.e.* C₆₊ alcohols and aldehydes) increase to a small extent over this range as condensation reactions between butanol and ethanol becomes more prevalent, though conversions remain too low for significant production of these species. The acetaldehyde yield is also fairly constant over this WHSV range, a reflection of the fact that it is produced at near equilibrium levels.

To examine the effect of water on the reaction, feeds containing water at levels between 0 and 11.3 wt% were examined at a fixed WHSV of 2.10 h⁻¹ (with respect to ethanol) as shown in Fig. 8. An inhibiting effect was observed with 4.1 wt% water in the feed, decreasing the conversion from 22.7 to 18.5%. This water content notably closely matches the

amount produced with a neat ethanol feed at this WHSV. Further increasing the water content to 11.3 wt% further decreased the conversion to 14.6%. Returning to the original conditions did not fully restore catalytic activity, thus irreversible deactivation may occur with high water contents (>4 wt%) beyond those present in the WHSV experiments where the effects of water on catalytic activity were reversible. To more quantifiably assess the inhibiting role of water, a rate order for water was approximated. The average partial pressure of water across the catalyst bed was estimated from the amount of water fed in and the expected amount of water produced based upon the other products observed (*e.g.* two moles of water are expected per mole of butene formed). The ethanol conversion rate was calculated as an average across the reactor based on the yield observed at the outlet. A log-log plot of the ethanol conversion rate *versus* the average water pressure (Fig. S4†) suggests that the rate order with respect to water is -0.23. While this is consistent with an inhibiting role of water, the value is not sufficiently negative to fully account for the minimal conversion increases observed with decreasing WHSV, suggesting that other species are likely inhibiting ethanol conversion as well. Carboxylate species derived from acetaldehyde or organic acids are also known to poison base sites.^{57,58} While organic acids could not be detected in this study, they may be formed at low levels *via* ester hydrolysis.¹¹ The combination of water and strongly-binding species produced either from ethanol coupling or through side-reactions are therefore likely responsible for decreasing catalytic activity at elevated conversions, which can make ethanol coupling at high conversions difficult to obtain with these catalysts.

Conclusions

AlMgO and AlCaO mixed oxides derived from hydrotalcite have been shown to be effective supports for the preparation of well-dispersed Cu catalysts with various concentrations of metal, acid, and base sites, as demonstrated by the measurements of NH₃- and CO₂-TPD and infrared with pyridine as probe molecule. Application of the emulsion-mediated preparation method altered the textural properties of the catalysts and improved the exposure of Cu on the surface of the supports. The beneficial impact of operating at elevated H₂ pressures with Cu/AlMgO (increased yield and selectivity to desired products) has been shown experimentally and linked to the shifting ethanol dehydrogenation equilibrium, which leads to undesirable high acetaldehyde selectivities at low pressures. At elevated pressures, Mg catalysts without Cu demonstrate moderate activity for ethanol conversion though Ca catalysts are essentially inactive. The addition of Cu promoted dehydrogenation for all catalysts and thereby increased the activity for ethanol conversion. The selectivity to higher alcohols and aldehydes increased from 30.7 to 54.8% upon addition of Cu to AlMgO while the conversion increased from 7.1 to 19.7%. Diethyl ether formation was suppressed, shifting unfavourable dehydration toward olefin

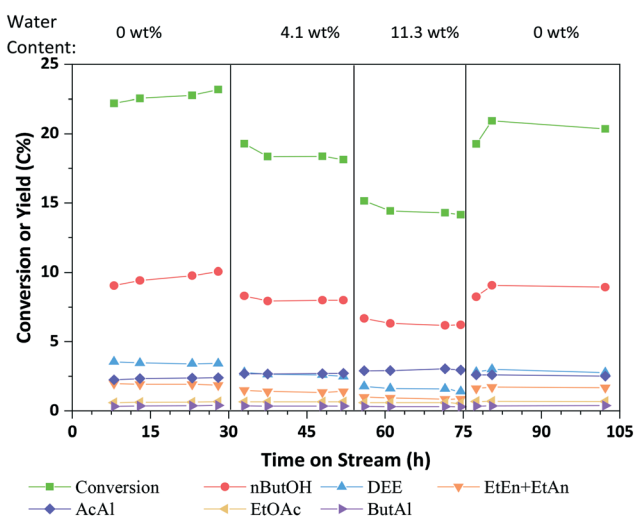


Fig. 8 Ethanol conversion over Cu/AlMgO-P with varying amounts of water added to the ethanol feed. Conditions: 325 °C, 675 mg catalyst, 30–33 μL min⁻¹ liquid feed (0–11.3 wt% water in ethanol), 86–90 mL min⁻¹ H₂, 32 bar total pressure.

and paraffin production. The porosity had a small effect on the conversion of ethanol with the observed trends correlating more clearly with surface area changes. Increasing space time by three times increased ethanol conversion over Cu/AlMgO-P by a factor of only 1.4 times, which is attributed to inhibition by species produced in the reaction.

Conflicts of interest

There are no conflicts to declare.

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